S. SASIKUMAR

AI DevOps Engineer | AWS DevOps

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PROFESSIONAL SUMMARY

AI DevOps Engineer hands-on DevOps and AWS cloud experience, specializing in deploying, scaling, and operating machine learning and generative AI workloads in production. Skilled in building MLOps, pipelines with Amazon SageMaker, MLflow, and Amazon Bedrock; serving models with FastAPI and Docker on Amazon EKS (GPU node groups, Helm, autoscaling); and automating infrastructure with Terraform and Ansible. Strong background in CI/CD (Jenkins, GitHub Actions), observability (CloudWatch, Prometheus, Grafana), DevSecOps, cost optimization of GPU/training workloads, and maintaining 99.9% availability in SLA-driven environments.

TECHNICAL SKILLS

AI / MLOps : Amazon SageMaker, Amazon Bedrock, MLflow, Model Training & Registry, Model Versioning, Model Serving (FastAPI, Docker), LLM Application Deployment, Automated Model Release & Rollback, AI-Assisted Automation, GitHub Copilot, LLM APIs, RAG Pipelines, GPU Inference Optimization

Cloud (AWS): EC2, VPC, S3, IAM, RDS, ECR, EKS, Route 53, ELB/ALB, Auto Scaling, CloudWatch, CloudTrail, Secrets Manager, KMS, Spot Instances

Containers & Orchestration: Docker, Kubernetes, Amazon EKS, Helm, GPU Node Groups, Horizontal Pod Autoscaler, IRSA

CI/CD: Jenkins, GitHub Actions, GitLab CI/CD, Blue-Green Deployment, Canary Deployment, Image Scanning

Infrastructure as Code: Terraform, AWS CloudFormation, Ansible, Reusable Modules

Monitoring & Observability: CloudWatch, Prometheus, Grafana, ELK Stack, Latency/Throughput/Error-Rate and CPU/GPU Utilization Monitoring

Security (DevSecOps): IAM Least Privilege, Secrets Manager, KMS Encryption, Container Vulnerability Scanning, CloudTrail Auditing

Scripting & OS: Python, Bash, PowerShell, Linux (RHEL/Ubuntu), Windows Server

Networking: VPC, Subnets, VPC Peering, NAT Gateway, Security Groups, DNS, DHCP, Load Balancing

Tools & Process: Git, GitHub, GitLab, ServiceNow, ITIL, Agile/Scrum, Incident Management, Cost Optimization

PROFESSIONAL EXPERIENCE

AI DevOps Engineer | Alfa Information Private Limited — 08/2025 – 09/2026

- Designed and automated CI/CD pipelines using Jenkins and GitHub Actions for AI/ML and LLM-based applications, covering build, test, container image scanning, and deployment across dev, staging, and production environments.
- Containerized machine learning models and inference APIs with Docker and deployed them on Amazon EKS using Helm, with Horizontal Pod Autoscaling and GPU-enabled node groups for model-serving workloads.
- Built MLOps workflows for model training, versioning, registry, and automated release using Amazon SageMaker and MLflow, including automated rollback of failed model deployments.
- Provisioned AWS infrastructure for AI workloads (VPC, EC2 GPU instances, S3, ECR, IAM, RDS) as reusable Terraform modules and managed configuration with Ansible.
- Integrated generative AI services (Amazon Bedrock and LLM APIs) into applications, securing credentials with AWS Secrets Manager, KMS encryption, and IAM Roles for Service Accounts (IRSA) under least-privilege access.
- Implemented monitoring and alerting with CloudWatch, Prometheus, and Grafana to track latency, throughput, error rates, and CPU/GPU utilization of AI services, improving incident detection and response times.
- Optimized cost and performance of AI workloads through right-sizing, Spot instances for training jobs, autoscaling, and scheduled shutdown of non-production environments.
- Embedded DevSecOps practices (container vulnerability scanning, secrets management, CloudTrail auditing); collaborated with data science and development teams to productionize models, documented runbooks, and handled incidents through ServiceNow.

AWS DevOps Engineer | **Crown Solutions Pvt Ltd** **02/2024 – 07/2025**

- Designed, deployed, and managed highly available AWS cloud infrastructure using core services (VPC, EC2, S3, RDS, IAM, Route 53, ALB, Auto Scaling) supporting mission-critical enterprise workloads.
- Built and optimized CI/CD pipelines with Jenkins and GitHub Actions, automating deployments across dev, staging, and production and significantly reducing release cycle time.
- Implemented Infrastructure as Code with Terraform to provision secure, repeatable multi-tier cloud architecture, reducing manual configuration errors.
- Containerized legacy enterprise applications with Docker and managed scaling and orchestration on Amazon EKS.
- Set up centralized monitoring, logging, and alerting with AWS CloudWatch, Prometheus, and Grafana to proactively identify and mitigate performance risks.
- Managed infrastructure change requests and incident tickets through ServiceNow, consistently meeting SLA commitments and maintaining 99.9% system availability.

DevOps / Cloud Infrastructure Engineer | **Madox Technologies** **05/2022 – 12/2023**

- Led cloud migrations from on-premises/hybrid environments to AWS with minimal disruption to data workflows and business operations.
- Automated configuration management, security patching, and package distribution across hundreds of EC2 instances using Ansible playbooks.
- Enforced least-privilege access across cloud resources through custom IAM policies, Security Groups, and Network ACLs.
- Built automated backup workflows for Amazon RDS and S3 to support cross-region disaster recovery.
- Developed Bash and PowerShell automation scripts that improved infrastructure operational efficiency by 25%.
- Maintained security posture through regular access audits and change tracking via AWS CloudTrail.

Systems & Network Administrator | **Quess Corp Ltd** **11/2020 – 11/2021**

- Managed and scaled Windows Server infrastructure and Active Directory environments for enterprise users.
- Administered Office 365 for 300+ users, handling provisioning, Exchange Online mailbox management, and licensing.
- Configured Group Policy Objects (GPOs) to strengthen security and compliance posture.
- Managed corporate network infrastructure including DHCP, DNS, Fortinet VPN, and routing diagnostics; supported hybrid on-premises-to-cloud connectivity.

KEY PROJECTS

Enterprise AI/MLOps & LLM Platform on AWS EKS *(Alfa Information Private Limited)*

Tech Stack: AWS (EKS, SageMaker, Bedrock, ECR, S3, IAM), Terraform, Kubernetes, Helm, Docker, GitHub Actions, MLflow, FastAPI, Prometheus, Grafana, GitHub Copilot, Python

- Architected a production-grade platform on AWS to train, version, deploy, and serve machine learning and generative AI (LLM) workloads, standardizing the path from model development to production for data science and application teams.
- Provisioned the entire platform as reusable, version-controlled Terraform modules (VPC, EKS, GPU node groups, ECR, S3, IAM, RDS) with remote state in S3 and DynamoDB locking, enabling repeatable, one-command environment builds across dev, staging, and production.
- Deployed containerized FastAPI model-serving and LLM inference APIs on Amazon EKS using Helm charts, with GPU node groups, Horizontal Pod Autoscaler, and Karpenter/Cluster Autoscaler to scale on request load while controlling GPU spend.
- Built GitHub Actions CI/CD pipelines (GitOps-style, pull-request driven) that run linting, unit tests, Terraform plan/apply, Docker image build, vulnerability scanning, and Helm deployment with environment approvals and automated rollback.
- Implemented an MLOps lifecycle with Amazon SageMaker and MLflow for experiment tracking, model registry, versioning, and automated promotion of validated models; integrated Amazon Bedrock and LLM APIs for generative AI features, including RAG-style application services.

- Used AI-assisted tooling (GitHub Copilot and LLM assistants) to accelerate Terraform, Helm, and pipeline authoring, generate test cases, and draft runbooks, while enforcing code review and policy checks on all AI-generated changes.
- Secured the platform with IAM Roles for Service Accounts (IRSA), AWS Secrets Manager, KMS encryption, private subnets, and image scanning, following least-privilege access and audit logging with CloudTrail.
- Set up observability with Prometheus, Grafana, and CloudWatch dashboards and alerts for inference latency, throughput, error rate, and CPU/GPU utilization; reduced idle GPU cost using Spot instances for training and scheduled scale-down of non-production clusters.

CI/CD & Cloud Infrastructure for a Financial Trading Platform *(Client: Confidential)*

- Designed an end-to-end CI/CD pipeline for a low-latency financial trading and market-data platform, automating build, test, and deployment across dev, UAT, and production using Jenkins and GitHub Actions.
- Provisioned highly available, fault-tolerant AWS infrastructure (multi-AZ VPC, EC2, Application/Network Load Balancer, RDS, S3) using Terraform to meet strict uptime and compliance requirements.
- Containerized trading and market-data microservices with Docker and orchestrated them on Amazon EKS with auto-scaling for peak market-hours load.
- Built automated security and compliance checks (IAM least-privilege, encrypted storage, CloudTrail audit logging) into the pipeline.
- Implemented blue-green deployment with automated rollback for zero-downtime releases during live trading hours; configured real-time monitoring with CloudWatch, Prometheus, and Grafana.

AWS Infrastructure Automation & Kubernetes Migration for Enterprise Application

- Migrated a monolithic enterprise application from on-premises servers to a fully automated, containerized AWS environment, improving deployment frequency and reducing infrastructure cost.
- Built modular, reusable Terraform code to provision VPC, EC2, RDS, S3, and IAM across dev, staging, and production accounts with a single source of truth.
- Re-architected the application into microservices on Docker and Amazon EKS, enabling independent scaling and faster rollbacks.
- Designed a multi-stage CI/CD pipeline (build, test, security scan, deploy) using Jenkins and GitHub Actions, cutting manual release effort and deployment errors.
- Implemented centralized logging and monitoring (CloudWatch, Prometheus, Grafana, ELK) with automated alerting to reduce MTTD and MTTR; achieved 99.9% uptime with auto-scaling, load balancing, and multi-AZ disaster recovery.

EDUCATION

Diploma in Electrical & Electronics Engineering (EEE) – Sri Venkatachalapathy Polytechnic College, Villupuram